aLab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

_______show ip route ospf____

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ____Serial0/0/0____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? __129____

Does R2 show up as an OSPF neighbor on R1? ___Yes___

Does R2 show up as an OSPF neighbor on R3? __No___

What does this information tell you?

___R2 has re-established an OSPF adjacency with R1 only. Since S0/0/1 on R2 is still passive, R3 does not form a neighbor relationship with R2 and must reach 192.168.2.0/24 indirectly through R1____

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

_______R2(config)# router ospf 1_______________________________________

________R2(config-router)# no passive-interface s0/0/1_______________________

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? ___S0/0/1___

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

____The accumulated cost is 65

It is calculated by adding the OSPF cost of R3's outgoing interface S0/0/1 (cost: 64) and the cost of R2's destination interface GigabitEthernet0/0 (cost: 1)

64+1=65

Is R2 listed as an OSPF neighbor to R3? _Yes___ 2

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

_____Because the default reference bandwidth is only 100Mbps, OSPF assigns a cost of 1 to all interfaces operating at 100Mbps or faster

Changing it to a higher value allows OSPF to accurately differentiate the costs and choose the best path among faster modern links. 2

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

__The default reference bandwidth (100Mbps) was used, and R1 serial interfaces were manually set to a bandwidth of 128 Kbps. The serial link cost became (100000000/128000) = 781

Gigabit interface cost is 1.

Cost to 192.168.3.0/24=R1 S0/0/1 cost 781+R3 G0/0 cost 1=782.

Cost to 192.168.23.0/30=R1 S0/0/0 cost 781+R2 S0/0/1 default serial cost 64= 845. 2

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

___The new accumulated cost is 1562.

This is because all serial interfaces in the topology have now been configured with a bandwidth of 128 Kbps, making each serial link's cost 781.

The total cost is R1serial interface cost 781+neighbor router serial interface cost 781=1562 2

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

___We manually configured the OSPF cost of R1 direct link to R3 S0/0/1 to 1565

The total cost for the direct path became 1566 (1565 + 1 for R3 G0/0)

However, the alternative path routing through R2 has a lower total cost of 1563 (781 for R1 to R2+781 for R2 to R3+1 for R3's G0/0)

OSPF automatically chose the path via R2 because it has the lowest accumulated cost. 4

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   ___It makes it much easier to identify each router when checking the routing table or troubleshooting.

   It also makes the OSPF process more stable and predictable 4

2. Why is the DR/BDR election process not a concern in this lab?

   Because the routers in this lab are connected using serial cables, which are point-to-point links.

   DR/BDR elections only happen on multi-access networks like Ethernet. 4

3. Why would you want to set an OSPF interface to passive?

   ___To stop sending OSPF Hello packets to the LAN where only PCs are connected.

   Since there are no other routers there,it saves bandwidth and is more secure.____________________________________ 4

   ___

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   ___OSPF uses "Areas" (like Area 0).

   Dividing a big network into smaller areas keeps routing tables small and reduces update traffic, which helps the network scale.____________________ 4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   _________network 10.0.0.0 0.255.255.255 area 0__________ 4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   ____________The EIGRP route will be used. When comparing different protocols, the router checks the AD, not the metric. EIGRP has the lowest AD (90) compared to OSPF (110) and RIP (120).__________________ 4